RAGNI **RUPAM**

Consultant, Senior Data Scientist

Bengaluru,Karnataka,India

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SUMMARY

Senior Data Scientist with 9+ years of experience delivering end-to-end data science and machine learning solutions across FinCrime, AML, Risk & Compliance, Healthcare, and Supply Chain domains. Proven track record of building and deploying classification, alert prioritization, anomaly detection, and optimization models using Python and SQL on large-scale enterprise data. Strong expertise in statistical analysis, NLP, and applied machine learning, with experience translating complex analytical outcomes into clear, actionable insights for business and regulatory stakeholders. Adept at client interaction, mentoring team members, and driving data-driven decision-making in consulting environments.

EDUCATION

B.Tech in Electronics & Communication Engineering
Maulana Abul Kalam Azad University of Technology (WBUT)
Kolkata, West Bengal
Year of Passing- 2016

MTech in Data Science and Engineering
Birla Institute Of Technology And Science – Pilani
(BITS–Pilani WILP)
Year of Passing- 2024

EXPERIENCE

Senior Data Scientist at Deloitte
Bengaluru, Karnataka

Data Scientist at Ernst & Young (EY)
Bengaluru, Karnataka

Senior Analyst-Data Science at Infosys
Bengaluru, Karnataka

System Engineer at Tata Consultancy Services
Noida, Uttar Pradesh

SKILLS

Programming Languages

- Python
- SQL

Machine Learning & Data Science

Data Cleaning, Feature Engineering, Exploratory Data Analysis (EDA), Classification, Regression (Linear & Logistic), Clustering, Time Series Analysis, Ensemble Models, Random Forest, XGBoost, SVM, K-Means, k-NN, Gradient Descent, Anomaly Detection, Model Evaluation & Validation

Tools & Frameworks

Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, NLTK, AWS, MLOps Concepts, Power BI, Microsoft Excel, Aera, OpenAI APIs

Statistics & Analytics

Hypothesis Testing, Regression Analysis, Principal Component Analysis (PCA), T-SNE, Statistical Modelling

Natural Language Processing & AI

Text Preprocessing, Bag of Words, TF-IDF, Word2Vec, NLP Pipelines, Large Language Models (LLMs), LangChain, Retrieval-Augmented Generation (RAG), Generative AI (applied use cases)

Domain Expertise

FinCrime Analytics, AML & Compliance Monitoring, Transaction Monitoring, Alert Prioritization, Threshold Tuning, Risk Analytics, Customer Segmentation, Supply Chain Analytics & Optimization

PROJECTS

Senior Data Scientist at Deloitte. *Jan 2024- present*

Bengaluru, Karnataka

- **Led end-to-end data migration and transformation pipelines**, extracting data using SQL and performing Python-based transformations to create high-quality, analytics-ready datasets for downstream ML and compliance use cases.
- **Led development of dynamic safety stock optimization solutions**, leveraging historical demand patterns and lead-time variability to support data-driven inventory planning and operational decision-making.

- **Applied machine learning–driven optimization using Aera**, enabling scenario-based analysis and supporting business-aligned recommendations within enterprise planning workflows.
- **Delivered a machine learning project**, applying supervised ML techniques on structured enterprise data to improve predictive accuracy and support operational and analytical decision-making.
- **Executed a data science project**, involving exploratory data analysis, feature engineering, and statistical validation to uncover patterns, improve data quality, and generate actionable business insights.
- **Led design and implementation of Python automation scripts** for data validation and log validation, significantly reducing manual effort and improving consistency across migration cycles.
- **Developed automated reporting pipelines** to process CSV-based system logs and generate structured PDF validation reports using Python, improving auditability and stakeholder communication.
- **Applied statistical validation and ML-based data quality checks** to ensure reliability and consistency of migrated datasets, enabling trusted downstream analytics and compliance reporting.
- **Collaborated closely with clients and business stakeholders**, translating analytical findings, validation outcomes, and optimization logic into actionable insights aligned with business and regulatory expectations.
- **Supervised and mentored team members** on data preparation, validation approaches, automation practices, and ML implementation standards, ensuring reproducibility and scalability of deliverables.
- **Built and integrated applied Generative AI solutions**, including document parsing, summarization, domain-specific question answering, and **agentic AI workflows**, to support intelligent automation and internal knowledge assistance.

Data Scientist at Ernst & Young (EY)

Apr 2022- Dec 2023

Bengaluru, Karnataka

- **Designed and developed a machine learning–based alert prioritization model** to categorize AML and compliance alerts into high, medium, and low risk, significantly improving investigator focus and case handling efficiency.
- **Owned end-to-end model development** for alert prioritization, including problem framing, data understanding, feature engineering, model training, evaluation, and iterative refinement using historical transaction and alert data.
- Applied **advanced feature engineering and statistical transformations** on large-scale financial datasets to capture risk signals, behavioral patterns, and transactional anomalies.
- Performed **extensive exploratory data analysis and visualization** to understand alert drivers, distribution shifts, and system-generated noise impacting alert volumes.
- Evaluated model performance using **classification and risk metrics** (such as precision, recall, and stability measures), ensuring robustness and scalability across high-volume transaction systems.
- Integrated alert prioritization outputs into **transaction monitoring workflows**, supporting more effective triaging of customer and non-customer alerts.
- Worked extensively on **AML and FinCrime systems**, analyzing transactions for fraud, compliance breaches, and regulatory risks in alignment with global standards.
- Conducted analysis on the **impact of inflation on threshold values**, assessing changes in transaction behavior and alert volumes across multiple lines of business and geographies.
- Collaborated with **cross-functional teams** including compliance, technology, and business stakeholders to align model logic with regulatory expectations and operational constraints.
- Presented **model methodology, results, and business impact** to senior stakeholders, translating complex analytical outcomes into actionable recommendations.
- Interacted directly with **clients**, gathering requirements, addressing feedback, and ensuring analytical solutions met both business objectives and regulatory needs.

Senior Analyst-Data Science at Infosys

Sep 2021- Apr 2022

Bengaluru, Karnataka

- Contributed to the development of a **lung cancer prediction model** using Python and SQL, leveraging large-scale insurance claims data to support product-specific market forecasting.
- Performed **in-depth data analysis and feature engineering** to improve model accuracy, stability, and relevance of predictive outputs.
- Worked closely with **cross-functional teams** to collect, clean, and curate insurance claims data, ensuring data completeness and analytical integrity.
- Implemented and tuned **machine learning algorithms** in Python to optimize model performance and scalability.
- Utilized **SQL-based data extraction and transformation** to prepare modelling datasets, enabling efficient and repeatable analytical workflows.
- Validated model performance through **rigorous testing and evaluation**, ensuring reliability of predictions for business use.
- Presented **analytical findings and model insights** to stakeholders, including product and leadership teams, to support informed decision-making.

- Applied structured **problem-solving and analytical approaches** to address data quality and modelling challenges throughout the project lifecycle.

System Engineer at Tata Consultancy Services

Aug 2016- Sept 2021

Noida, Uttar Pradesh

- Developed a **malware class prediction system** using Python and SQL to classify data into multiple malware categories, strengthening early exposure to supervised machine learning and pattern recognition problems.
- Designed and implemented a **product recommendation system** leveraging product features and similarity-based techniques to generate personalized recommendations.
- Worked extensively on **Oracle E-Business Suite (EBS) production support and development**, handling complex business logic through packages, procedures, functions, views, and cursors.
- Contributed to **Oracle application development activities**, including concurrent requests, request sets, value sets, profiles, data loading, reconciliation, and application configurations.
- Automated recurring operational tasks and validation steps, improving process efficiency and reducing manual intervention across support workflows.
- Collaborated with cross-functional technical and business teams to ensure **stable production operations and timely delivery** of client requirements.
- Built a strong foundation in **Python, SQL, and enterprise systems**, which later enabled a transition into data science and machine learning roles.

LANGUAGES

- Hindi
- English